HapticNV MCU Software Porting Guide

1 Software Overview

Driver File	haptic_nv.c,haptic_nv.h,haptic_nv_reg.h,haptic_nv_params.c,haptic_nv_config.c, aw862x.c,aw862xx.c,aw8623x.c
IC Support	AW862X:aw8623 aw8624 AW862XX:aw86214 aw86223 aw86224 aw86225 AW8623X:aw86233 aw86234 aw86235 AW8624X:aw86243 aw86245
I²C Address	0x5A/0x58

2 Software Porting

1. Open the haptic_nv_config.c file in the driver package and implement the functions in the file based on your own platform
2. Open the driver package aw862x.c/aw862xx.c/aw8623x.c/aw8624x.c file, using the motor nominal f0 and rated voltage according to modify the dts f0_pre and cont_lra_vrms parameters of the structure
3. Put all *.c and *. h files into the Awinic folder in the project directory
4. Right-click "Project", create "Awinic" module, add aw862x.c, aw862xx.c, aw8623x.c, aw8624x.c haptic_nv.c，haptic_nv_config.c，haptic_nv_params.c
5. Open haptic_hv.h, select the chip macro control AW862X_DRIVER/ AW862XX_DRIVER/ AW8623X_DRIVER/ AW8624X_DRIVER according to the chip used, reduce the amount of code, and save platform resources
6. Open haptic_nv.h, and enable and disable AW_RTP_AUTOSIN macro control according to whether RTP autosin function is used (disabled by default).
7. Open haptic_nv.h, AW_IRQ_CONFIG config using interrupt, otherwise use trig function, AW_CHECK_RAM_DATA config ram read back check, AW_ENABLE_RTP_PRINT_LOG open rtp play process detailed log, HAPTIC_NV_DOUBLE is configured for use with dual motors
8. Add #include "haptic_nv.h" to main.c, configure the I2C and timer (hi2c1 , hi2c2 and htim3 are used by default for the driver), and compile the project

3 Validation Of Driver Migration

Chip identification is successful:

```
[haptic_nv][0749]haptic_nv_boot_init: haptic nv driver version v0.2.0
[haptic_nv][0476]parse_chipid: detected aw8624.
[haptic_nv][0442]sw_reset: enter!
```

```
[haptic_nv][0749]haptic_nv_boot_init: haptic_nv driver version v0.2.0
[haptic_nv][0091]haptic_nv_read_chipid: reading chip id
[haptic_nv][0091]haptic_nv_read_chipid: reading chip id
[haptic_nv][0091]haptic_nv_read_chipid: reading chip id
[haptic_nv][0091]haptic_nv_read_chipid: reading chip id
[haptic_nv][0091]haptic_nv_read_chipid: reading chip id
[haptic_nv][0466]parse_chipid: try to replace i2c addr [(0x58)] to read chip id again
[haptic_nv][0487]parse_chipid: aw86224 or aw86225 detected
[haptic_nv][0442]sw_reset: enter!
```

```
[haptic_nv][0582]parse_chipid: aw86233 detected
[haptic_nv][0521]sw_reset: enter!
[haptic_nv][0768]aw8623x_interrupt_setup: enter
```

```
[I][haptic_nv]parse_chipid: try to replace i2c addr [(0x58)] to read chip id again
[I][haptic_nv]parse_chipid: aw86243 detected
```

4 Function And Interfaces

The Haptic_NV MCU compatible driver defines a function pointer structure. After calling the initialization function haptic_nv_boot_init to complete startup initialization, the user can call the function interface through the structure pointer g_func_haptic_nv, as follows：

```
struct aw_haptic_func
{
int (*check_qualify)(void);                          //Chip mass production verification
int (*get_irq_state)(void);                          //Get the interrupt status
int (*get_f0)(void);                                 //Get motor frequency f0
int (*offset_cali)(void);                            //Start offset calibration
void (*trig_init)(void);                             //trig initialization
void (*irq_clear)(void);                             //Read clear interrupt
void (*haptic_start)(void);                          //Start playing
void (*play_stop)(void);                             //Stop playing
void (*cont_config)(void);                           //Register configuration in conT mode
void (*play_mode)(uint8_t);                          //Select the playback mode
void (*ram_init)(aw_bool);                           //RAMINIT enable control
void (*misc_para_init)(void);                        //Initialize the boot register
void (*interrupt_setup)(void);                       //Interrupt the configuration
void (*vbat_mode_config)(uint8_t);                   //In vbat mode
void (*protect_config)(uint8_t, uint8_t);            //Motor protection configuration
void (*calculate_cali_data)(void);                   //Calculate the f0 calibration value
void (*set_gain)(uint8_t);                           //Set the gain
void (*set_wav_seq)(uint8_t, uint8_t);               //Set the waveform
void (*set_wav_loop)(uint8_t, uint8_t);              //Set the playback times
void (*set_rtp_autosin)(uint8_t);                    //Set rtp autosin enable switch
void (*set_rtp_data)(uint8_t *, uint32_t);           //Write RTP data
void (*set_fifo_addr)(void);                         //Set the FIFO address
void (*get_fifo_addr)(void);                         //Get the FIFO address
void (*set_ram_data)(uint8_t *, int);                //Write ram data
void (*get_ram_data)(uint8_t *, int);                //Read ram data
void (*set_ram_addr)(void);                          //Set the RAM address
void (*set_repeat_seq)(uint8_t);                     //Set waveform playback to repeat
void (*set_base_addr)(void);                         //Set the base address
void (*set_trim_lra)(uint8_t);                       //Set the TRIM_LRA register
void (*set_rtp_aei)(aw_bool);                        //Set fast air break
void (*get_vbat)(void);                              //Check the battery voltage
void (*get_reg)(void);                               //Get register data
void (*get_lra_resistance)(void);                    //Motor impedance detection
uint8_t (*get_glb_state)(void);                      //Get the value of the global status register
uint8_t (*judge_rtp_going)(void);                    //Check whether RTP is being played
uint8_t (*rtp_get_fifo_afs)(void);                   //Check whether the FIFO is full
void (*f0_show)(void);                               //Get motor real F0
void (*ram_show)(void);                              //Read back to display RAM waveform data
```

```
void (*cali_show)(void);                                    //Get the calibrated motor F0
void (*irq_handle)(void);                                   //Interrupt handler function
void (*get_ram_num)(void);                                  //Get the number of RAM waveforms
void (*rtp_vib_work)( uint8_t);                             //RTP playback sample function
void (*set_hw_irq_status)(uint8_t);                         //Set the interrupt state
uint8_t (*get_hw_irq_status)(void);                         //Get the interrupt status
int (*f0_cali)(void);                                       //F0 calibration
int (*rtp_going)(void);                                     //RTP play
int (*long_vib_work)(uint8_t, uint32_t);                    //Long shock playback example function
int (*short_vib_work)(uint8_t, uint8_t, uint8_t);           //Short shock playback example function
int (*dual_short_vib)(uint8_t, uint8_t, uint8_t, uint8_t, uint8_t, uint8_t);// Double motor short shock
int (*dual_long_vib)(uint8_t, uint8_t, uint8_t, uint8_t, uint32_t);// Double motor long shock
};
```

5 Single Motor Function Call Reference

1. Initialization process:

Call the initialization function haptic_nv_boot_init() to initialize the haptic chip. If 0 is returned, the initialization is successful. Other interfaces can be used only after the initialization is complete.

2. Short vibration call example:
   g_func_haptic_nv ->short_vib_work(1, 0x80, 1);
3. Long vibration call example:
   g_func_haptic_nv ->long_vib_work(4, 0x80 , AW_MS_UINT*500);
4. RTP playback example:
   g_func_haptic_nv ->rtp_vib_work(0x80);
5. F0 calibration:
   g_func_haptic_nv ->f0_cali();
6. F0 reading after calibration:
   g_func_haptic_nv ->cali_show();
7. Real F0 read:
   g_func_haptic_nv ->f0_show();
8. Motor impedance detection:
   g_func_haptic_nv ->get_lra_resistance();
9. Battery voltage detection:
   g_func_haptic_nv ->get_vbat();

6 Dual Motor Function Call Reference

1. Initialization process:

```
haptic_nv_change_motor(MOTOR_L);                    //Switch control left motor
haptic_nv_boot_init();                              //Left motor chip initialization
haptic_nv_change_motor(MOTOR_R);                    //Switch control right motor
haptic_nv_boot_init();                              //Right motor chip initialization
```

2. Short vibration call example:

```
g_func_haptic_nv->dual_short_vib(1 , 0x80 , 1 , 1 , 0x80 , 1);      //Short vibration of double motors
```

3. Long vibration call example:

```
g_func_haptic_nv->dual_long_vib(4 , 0x80 , 4 , 0x80 , AW_MS_UINT*500);  //Long vibration of double motors
```

4. F0 calibration:

```
haptic_nv_change_motor(MOTOR_L);                    // Switch control left motor
g_func_haptic_nv->f0_cali();                        //Left motor F0 calibration
haptic_nv_change_motor(MOTOR_R);                    // Switch control right motor
g_func_haptic_nv->f0_cali();                        // Right motor F0 calibration
```

5. F0 reading after calibration:

```
haptic_nv_change_motor(MOTOR_L);                    // Switch control left motor
g_func_haptic_nv->cali_show();                      //Left motor F0 after calibration
haptic_nv_change_motor(MOTOR_R);                    // Switch control right motor
g_func_haptic_nv->cali_show();                      // Right motor F0 after calibration
```

6. Real F0 read:

```
haptic_nv_change_motor(MOTOR_L);                    // Switch control left motor
g_func_haptic_nv->f0_show();                        // Left motor real F0
haptic_nv_change_motor(MOTOR_R);                    // Switch control right motor
g_func_haptic_nv->f0_show();                        // Right motor real F0
```

7. Motor impedance detection:

```
haptic_nv_change_motor(MOTOR_L);                    // Switch control left motor
g_func_haptic_nv-> get_lra_resistance ();           // Left motor impedance detection
haptic_nv_change_motor(MOTOR_R);                    // Switch control right motor
g_func_haptic_nv-> get_lra_resistance ();           // Right motor impedance detection
```

8. Battery voltage detection:

```
g haptic_nv_change_motor(MOTOR_L);                  // Switch control left motor
g_func_haptic_nv-> get_vbat();                      // Left motor voltage detection
haptic_nv_change_motor(MOTOR_R);                    // Switch control right motor
g_func_haptic_nv-> get_vbat();                      // Right motor voltage detection
```

7 F0 Calibration Instructions

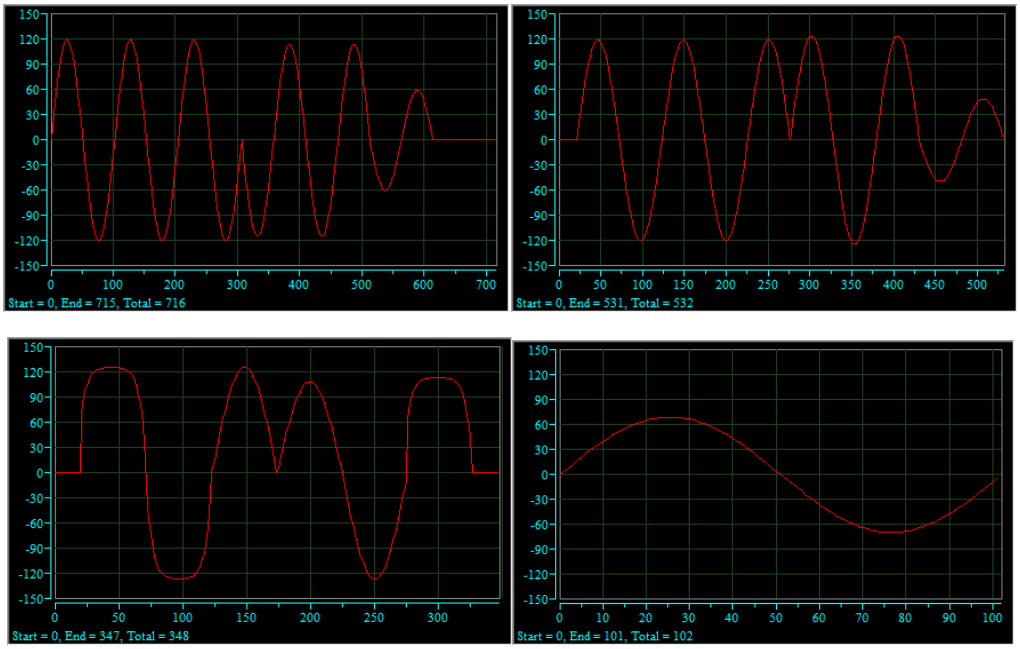
Two interfaces for saving and obtaining F0 calibration values are reserved in the haptic_hv_config.c, which need to be implemented by the customer:

1. void haptic_nv_set_cali_to_flash(void) // Write the calibration value into flash to save
2. void haptic_nv_get_cali_from_flash(void) // Read the calibration value from flash and update it to the driver

If the AW_F0_CALI_DURING_STARTUP macro is defined, the default device performs F0 calibration during startup, and the above two interfaces will not be used

8 Waveform Examples

Reference RAM waveforms(Wave1 ~ Wave3 are short seismic waveforms and Wave4 is long seismic specified waveforms.):



9 Precautions

1. RTP playback needs to be configured with external interrupt IRQ_PIN and falling edge trigger mode: when detecting hardware interrupt trigger, jump interrupt function processing;
2. In RTP mode, the i2c must be used independently by Haptic;
3. The driver needs to invoke the haptic_nv_boot_init function to initialize the g_func_haptic_nv pointer;
4. RTP is not supported by dual motor drive at present.

Revision History

Version	Date	Description
V1.0	2021-11-22	Initial release
V1.1	2022-01-27	1. Add the description of haptic_nv_config.c and improved driver adaptation steps
V1.2	2022-06-06	1. Key information is marked in red 2. Add migration validation 3. Add a watermark
V1.3	2022-08-05	1. Add aw8623x 2. Delete get_first_addr interface 3. Delete AW_LOG_CTRL
V1.4	2022-08-18	1. Add dual motor support
V1.5	2023-02-14	1. Delete superfluous *.h 2. Replacing a document template
V1.6	2024-02-05	1. Add AW_RTP_AUTOSIN macro control description. 2. Add aw8624x support.

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